Chapter 8

Market Relevance of Emerging Risk Measures

The previous chapter examined Cyber Risk in greater detail to prepare the financial analysis developed here. This chapter investigates whether news-based measures of cyber risk contain information about subsequent weekly equity portfolio excess returns. Coverage, Novelty, and Semantic Drift are examined through weekly regressions with six lags, following the empirical approach of Cherubini et al. Burst Detection is then used to examine periods of unusually intense media attention.

The analysis considers two equally weighted portfolios. The Diversified portfolio contains 195 firms from several industries. The Attacked + Security portfolio contains 76 firms: 66 companies classified as having experienced cyberattacks and 10 cybersecurity firms. The comparison allows the same news indicators to be evaluated for a broad collection of firms and for companies more directly connected to cybersecurity.

The empirical analysis begins with individual and combined tests of the news indicators. It then examines the contribution of Coverage to model fit and one-week-ahead forecasts. The final section studies whether the estimated relationships differ during Burst episodes and whether the Coverage results persist when contemporaneous Burst weeks are excluded.

8.1 Empirical Methodology

8.1.1 VAR specification

Let $R_{P,t}$ denote the weekly excess return of portfolio P in week t. The financial variable enters the regressions in levels. For each news indicator $I_t \in \{Coverage_t, Novelty_t, Drift_t\}$, the weekly change is defined as

$$\Delta I_t = I_t - I_{t-1}. \tag{8.1}$$

Weekly portfolio returns are already expressed as changes in asset value and need no additional differencing for the analysis developed here. Using $R_{P,t}$ also makes the estimated association directly interpretable in terms of subsequent weekly excess returns. The news variables enter in first differences to capture changes in reporting intensity and content. The six-lag structure follows the empirical framework used by Cherubini et al.; the present return equation is estimated with returns in levels.

For each portfolio and news indicator, the baseline bivariate VAR(6) is

$$\mathbf{Y}_{P,t} = \mathbf{c}_P + \sum_{\ell=1}^{6} \mathbf{A}_{P,\ell}\mathbf{Y}_{P,t-\ell} + \mathbf{u}_{P,t}, \qquad \mathbf{Y}_{P,t} = \begin{bmatrix} R_{P,t} \\ \Delta I_t \end{bmatrix}. \tag{8.2}$$

The financial equation used to assess the news indicator is

$$R_{P,t} = a_P + \sum_{\ell=1}^{6} b_{P,\ell} R_{P,t-\ell} + \sum_{\ell=1}^{6} c_{P,\ell}\Delta I_{t-\ell} + \varepsilon_{P,t}. \tag{8.3}$$

Each $c_{P,\ell}$ measures the conditional association between a change in the news indicator ℓ weeks earlier and the current weekly portfolio excess return. The coefficients are regression parameters from the return equation; they are not impulse responses to structural news shocks.

8.1.2 Tests and forecasting design

The null hypothesis for an individual indicator is

$$H_0 : c_{P,1} = \cdots = c_{P,6} = 0. \tag{8.4}$$

Joint restrictions are assessed with Wald tests based on heteroskedasticity- and autocorrelation-consistent (HAC) covariance estimates with a six-week bandwidth. The same procedure is applied to the relevant sets of lag coefficients in models containing multiple indicators. HAC(12) estimates provide a bandwidth sensitivity check for the individual and conditional results. Coefficients are estimated by ordinary least squares; HAC estimation affects their reported uncertainty.

Rejection of a joint restriction indicates that lagged changes in the relevant indicator are associated with weekly excess returns after controlling for lagged returns and any other included indicators. These tests concern the conditional relationship in the estimation sample. A separate expanding-window exercise evaluates one-week-ahead forecasts from an AR(6) benchmark and an augmented model containing six lags of $\Delta Coverage$. Forecast errors are compared using RMSE and out-of-sample R^2. The Clark–West test assesses the adjusted forecast-loss differential for the nested models, using one-sided HAC(6) inference.

8.2 Data and Portfolio Construction

8.2.1 News indicators and financial series

Coverage measures the share of news devoted to cyber risk. Novelty captures the arrival of less familiar information, while Semantic Drift tracks changes in the semantic representation of the topic. The measures are constructed from the same underlying news flow and describe different features of that flow.

The Diversified portfolio contains 195 companies from industries that include information technology, banking, and pharmaceuticals. The Attacked + Security portfolio contains 66 firms classified as Attacked and 10 classified as Security, following the grouping used by Cherubini et al. For each week,

the available valid stock returns are averaged with equal weights and the weekly approximation of the risk-free rate is subtracted. The resulting portfolio excess returns enter the financial equations as $R_{P,t}$.

8.2.2 Estimation sample

Figures 8.1–8.3 display the news indicators alongside the Diversified portfolio excess returns. Figures 8.4–8.6 present the corresponding series for Attacked + Security.

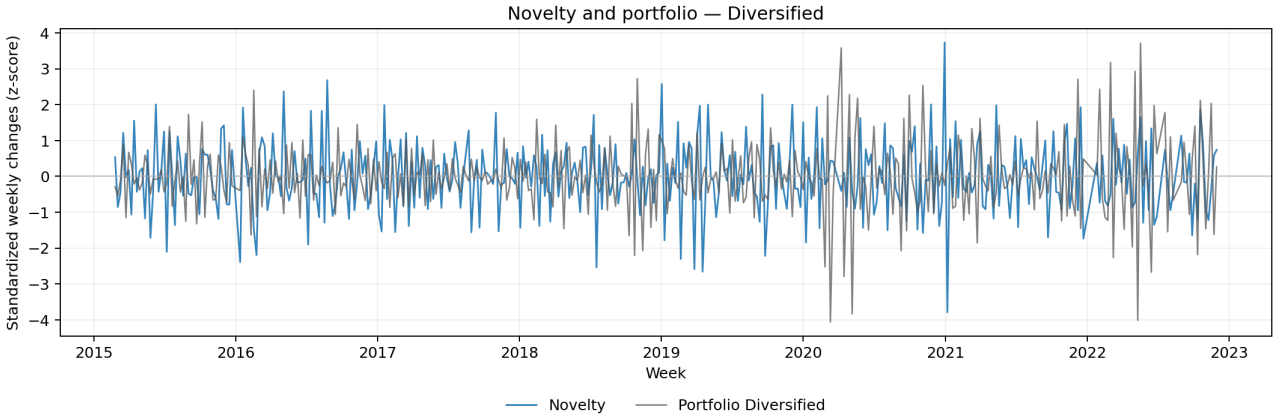


Figure 8.1: Weekly Novelty and Diversified portfolio excess returns over the observation period.

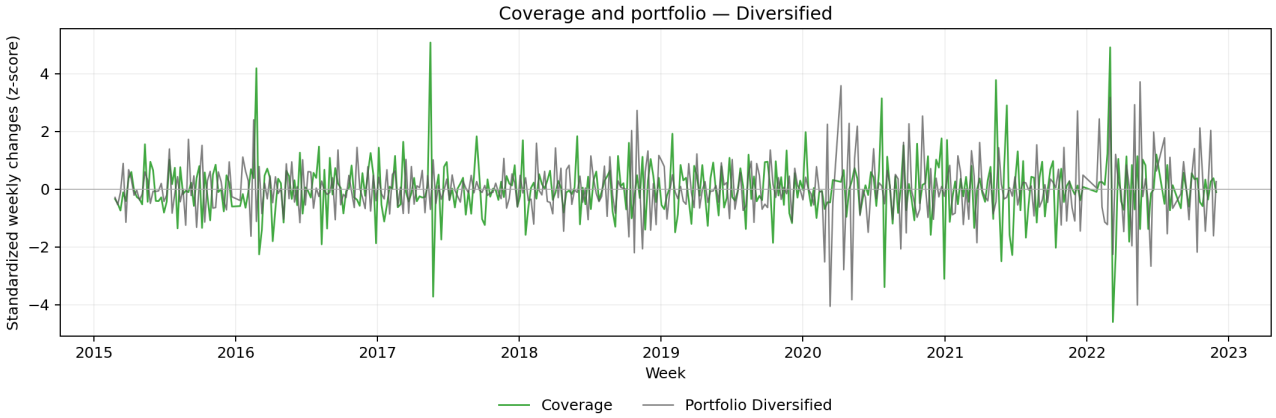


Figure 8.2: Weekly Coverage and Diversified portfolio excess returns over the observation period.

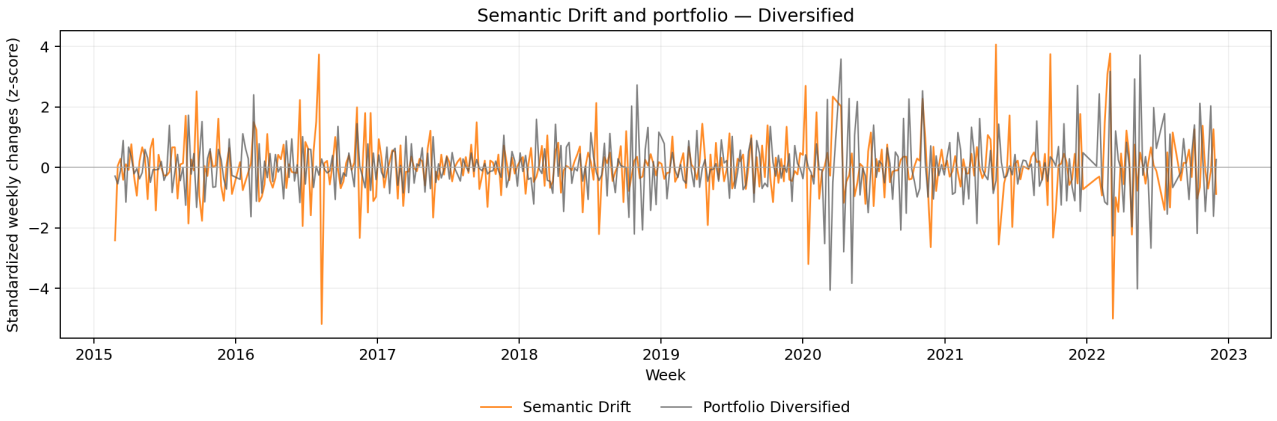


Figure 8.3: Weekly Semantic Drift and Diversified portfolio excess returns over the observation period.

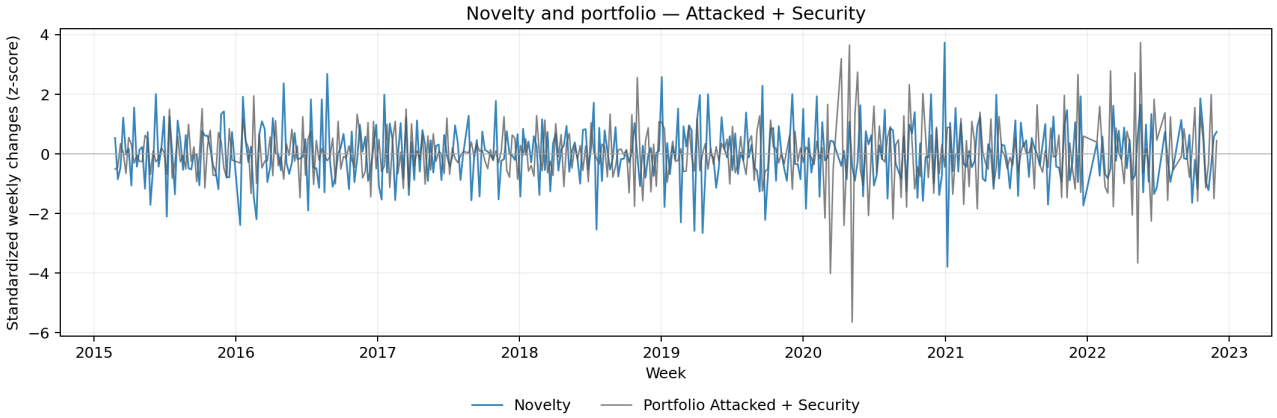


Figure 8.4: Weekly Novelty and Attacked + Security portfolio excess returns over the observation period.

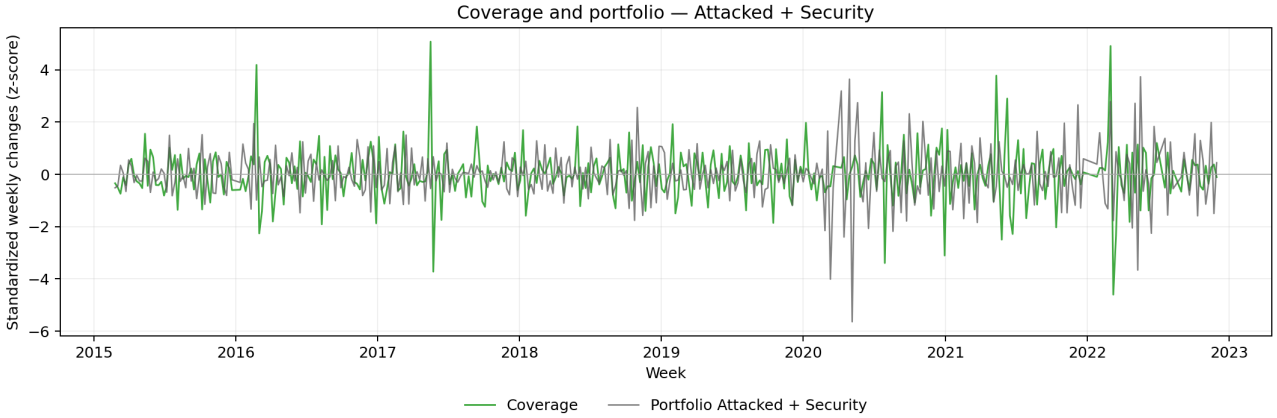


Figure 8.5: Weekly Coverage and Attacked + Security portfolio excess returns over the observation period.

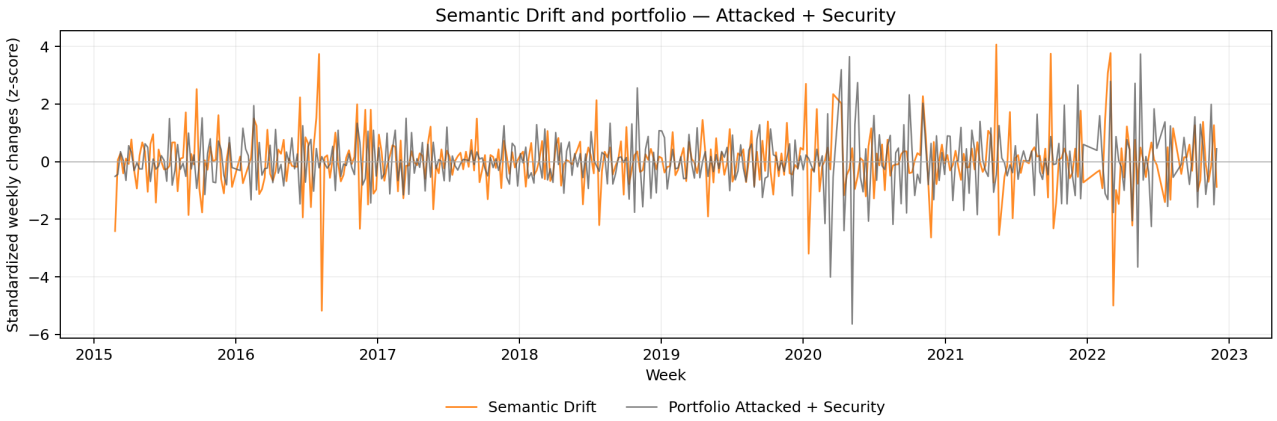


Figure 8.6: Weekly Semantic Drift and Attacked + Security portfolio excess returns over the observation period.

The plots describe the observed series. The regression tests use a common sample of 366 weekly observations from 25 February 2015 to 30 November 2022. This sample requires valid values for all three indicators and their lags, together with the financial variables. Lagged regressors are constructed on the original weekly calendar before the common sample is selected, preserving the intended weekly spacing across gaps in the available observations.
The Coverage forecasting exercise uses a dedicated sample of 406 consecutive valid weekly observations over the same calendar span. It retains dates that are unavailable in the common sample because of the additional indicator requirements. Its larger size therefore reflects a different availability restriction rather than a longer observation period.

8.3 Financial Relevance of News Indicators

8.3.1 Individual indicators

Table 8.1 reports conventional joint tests and HAC(6) tests for the six lagged coefficients of each indicator. All regressions use the same 366 observations.

Table 8.1: Individual News Indicators: Joint Tests of Six Weekly Lags

	Diversified		Attacked + Security	
Indicator	Classical	HAC(6)	Classical	HAC(6)
Coverage	0.0039	**0.0076**	0.0009	**0.0012**
Novelty	0.6389	0.6219	0.3790	0.3198
Semantic Drift	0.8538	0.4201	0.6935	0.2175

Notes: Entries are p-values for the joint restriction that all six lagged coefficients of the named indicator equal zero in the $R_{P,t}$ equation. Classical refers to the conventional joint F-test. HAC(6) uses a six-week covariance bandwidth. Bold denotes significance at the 5% level under HAC(6).

Coverage is significant at the 1% level in both portfolios under HAC(6). Novelty and Semantic Drift have p-values above 0.05 in both cases. Coverage also remains significant when the HAC bandwidth is extended to twelve weeks: the corresponding p-values are 0.0022 for Diversified and 0.0003 for Attacked + Security. Neither Novelty nor Semantic Drift reaches the 5% threshold under HAC(12)

(Novelty: 0.5565 and 0.2402; Semantic Drift: 0.2400 and 0.1202, for Diversified and Attacked + Security, respectively).
The evidence therefore points to Coverage as the strongest individual predictor among the three measures in these weekly return equations.

8.3.2 Combined indicators

Table 8.2 extends the tests to models containing two or three news indicators. Each indicator enters with six weekly lags, and the joint restriction covers all included news-indicator coefficients. The table reports HAC(6) inference.

Table 8.2: Combined News Indicators: HAC(6) Joint Tests

Model	Diversified	Attacked + Security
Coverage + Novelty	0.0931	**0.0154**
Coverage + Semantic Drift	**0.0394**	**0.0085**
Novelty + Semantic Drift	0.5514	0.2201
Coverage + Novelty + Semantic Drift	0.1751	**0.0334**

Notes: Entries are HAC(6) joint-test p-values for all news-indicator lag coefficients in the $R_{P,t}$ equation. Each included indicator contributes six lags. All specifications use 366 observations. Bold denotes significance at the 5% level.

Coverage and Semantic Drift are jointly significant in both portfolios. Coverage and Novelty are jointly significant in Attacked + Security, where the specification containing all three indicators is also significant. The Novelty–Semantic Drift combination is not significant in either portfolio. The combined tests impose more restrictions as indicators are added. For Diversified, Coverage is significant individually ($p = 0.0076$), while the Coverage–Novelty combination has $p = 0.0931$. Adding coefficients with little detectable incremental contribution can reduce the power of a joint test. The larger joint p-value therefore does not overturn the individual Coverage result. The conditional tests below examine whether each indicator contributes separately when the others are included.

Does Coverage retain its contribution when other indicators are included?

The full three-indicator model is jointly significant for Attacked + Security ($p = 0.0334$). Table 8.3 tests each indicator's six lag coefficients while retaining the other indicators and six lags of weekly excess returns.

Table 8.3: Conditional Tests in the Full Three-Indicator VAR(6)

Restriction tested	Diversified	Attacked + Security
Coverage lags = 0, given Novelty and Drift	**0.0220**	**0.0082**
Novelty lags = 0, given Coverage and Drift	0.9405	0.8107
Drift lags = 0, given Coverage and Novelty	0.6844	0.7111
Novelty and Drift lags = 0, given Coverage	0.9421	0.8536

Notes: Entries are HAC(6) joint-test p-values. All regressions include six lags of $R_{P,t}$ and the six lags of each news indicator. The final row jointly tests twelve coefficients. Each regression uses 366 observations. Bold denotes significance at the 5% level.

Coverage remains significant after controlling for Novelty and Semantic Drift, with p-values of 0.0220 for Diversified and 0.0082 for Attacked + Security. The joint test of Novelty and Semantic Drift, conditional on Coverage, yields 0.9421 and 0.8536. In the Attacked + Security portfolio, the significant full-model test is accompanied by a significant conditional contribution from Coverage.
With HAC(12), Coverage remains significant ($p = 0.0296$ and $p = 0.0195$). The joint p-values for Novelty and Semantic Drift, conditional on Coverage, are 0.9243 and 0.7532.
Table 8.4 examines the reduced model containing Coverage and Semantic Drift. Each indicator enters with six weekly lags, alongside six lags of portfolio excess returns.

Table 8.4: Conditional Tests in the Coverage–Semantic Drift VAR(6)

	Diversified		Attacked + Security	
Restriction tested	Classical	HAC(6)	Classical	HAC(6)
Coverage lags = 0, given Drift	0.0024	**0.0103**	0.0011	**0.0036**
Drift lags = 0, given Coverage	0.6599	0.6790	0.7010	0.7316

Notes: Entries are joint-test p-values for the six lags of the named indicator, conditional on the other news indicator and six lags of $R_{P,t}$. Each regression uses 366 observations. Bold denotes significance at the 5% level under HAC(6).

Coverage remains significant under HAC(6), with p-values of 0.0103 and 0.0036. Semantic Drift has conditional p-values of 0.6790 and 0.7316. With HAC(12), Coverage has p-values of 0.0072 and 0.0052, compared with 0.6700 and 0.6474 for Semantic Drift. The following section examines the lag coefficients of Coverage and its contribution to model fit.

8.4 Financial Information Content of Coverage

Table 8.5 presents the estimated Coverage coefficients in the individual bivariate return equations. Individual p-values use HAC(6) standard errors.

Table 8.5: Coverage Coefficients by Weekly Lag: HAC(6) Inference

	Diversified		Attacked + Security	
Lag	Coefficient	p-value	Coefficient	p-value
1	0.8332	**0.0345**	0.9476	**0.0162**
2	0.4056	0.2459	0.5349	0.1212
3	1.1545	**0.0068**	1.3566	**0.0030**
4	0.5383	0.0631	0.5557	**0.0432**
5	0.3157	0.2523	0.2651	0.3525
6	0.2017	0.4527	0.1112	0.7031

Notes: Coefficients are estimated by OLS in the $R_{P,t}$ equation; p-values use HAC(6) standard errors. The regressors are weekly changes in Coverage at the indicated lag. Bold denotes significance at the 5% level. Coefficients are conditional regression parameters.

All six estimated coefficients are positive in both portfolios. The first and third lags are significant in each portfolio, while the fourth lag is also significant for Attacked + Security. The largest coefficients occur at lag three, reaching 1.1545 for Diversified and 1.3566 for Attacked + Security.
A positive coefficient indicates that a rise in Coverage relative to the preceding week is associated with a higher subsequent weekly excess return, conditional on the included lags. The estimated scale

depends on the units of Coverage. The regressions do not identify the market mechanism behind this association, and the lag coefficients do not trace the full dynamic response to a structural news shock.

8.4.1 How much explanatory information does Coverage add?

To measure the contribution of Coverage, a restricted AR(6) return equation is compared with an augmented equation containing six additional lags of $\Delta Coverage$. Both models are fitted to the same 366 observations. The partial R^2 is

$$R^2_{\text{partial},P} = \frac{SSR_{\text{restricted},P} - SSR_{\text{Coverage},P}}{SSR_{\text{restricted},P}}, \tag{8.5}$$

where SSR denotes the residual sum of squares.

Table 8.6: Incremental In-Sample Explanatory Contribution of Coverage

Measure	Diversified	Attacked + Security
R^2, financial lags only	0.0177	0.0475
R^2, with Coverage	0.0693	0.1063
Increase in R^2 (percentage points)	5.16	5.88
Adjusted R^2, financial lags only	0.0013	0.0316
Adjusted R^2, with Coverage	0.0377	0.0759
Increase in adjusted R^2 (percentage points)	3.64	4.43
Partial R^2 of Coverage (%)	5.26	6.17
RMSE, financial lags only	0.02565	0.02750
RMSE, with Coverage	0.02497	0.02664
Reduction in RMSE (%)	2.66	3.14

Notes: Both equations contain an intercept and use 366 weekly observations. The restricted equation includes six lags of $R_{P,t}$; the augmented equation adds six lags of $\Delta Coverage$. RMSE is the square root of the mean squared fitted residual. Percentage-point changes in R^2 and percentage changes in RMSE are reported separately.

Adding Coverage raises adjusted R^2 by 3.64 percentage points for Diversified and 4.43 percentage points for Attacked + Security. The partial R^2 values indicate that the six Coverage lags remove 5.26% and 6.17% of the residual sum of squares remaining in the restricted financial equations. In-sample RMSE declines by 2.66% and 3.14%, respectively.

The relatively low R^2 of the restricted AR(6) equations reflects the limited linear dependence in weekly excess returns. The additional fit obtained with Coverage is measured against this benchmark on the same observations.

8.4.2 Financial interpretation of Coverage

Coverage measures the intensity of reporting about cyber risk. Its lagged weekly changes are positively associated with subsequent portfolio excess returns in both sets of firms. The association persists after the autoregressive return terms are included and after Semantic Drift is added to the equation.

The positive sign is consistent with several possible channels. Greater reporting may coincide with the diffusion of information about firms' ability to manage cyber incidents, or with broader developments that affect both news attention and equity markets. The available regressions do not distinguish these mechanisms. The observed coefficient pattern describes a conditional predictive association.

The fit comparison provides a measure of the association within the estimation sample. Forecast performance is examined separately because the regression coefficients must be re-estimated using information available before each forecasted week.

8.4.3 Out-of-Sample Forecasting Performance

The forecasting exercise compares an autoregressive benchmark with a model augmented by lagged Coverage. The benchmark is

$$R_{P,t} = \alpha_P + \sum_{k=1}^{6} \phi_{P,k} R_{P,t-k} + \varepsilon_{P,t}, \tag{8.6}$$

and the augmented model is

$$R_{P,t} = \alpha_P + \sum_{k=1}^{6} \phi_{P,k} R_{P,t-k} + \sum_{k=1}^{6} \beta_{P,k} \Delta Coverage_{t-k} + \varepsilon_{P,t}. \tag{8.7}$$

The dedicated Coverage sample contains 406 consecutive weekly observations. Both models are first estimated on 234 weeks. The estimation window then expands one week at a time, with coefficients fitted exclusively to observations preceding the forecasted week. This produces 172 one-week-ahead forecasts per portfolio from 21 August 2019 to 30 November 2022. No full-sample standardization of Coverage is performed within the forecasting procedure.
Forecast performance is summarized by RMSE and

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\text{OOS}}} (R_{P,t} - \widehat{R}^{Coverage}_{P,t})^2}{\sum_{t \in \mathcal{T}_{\text{OOS}}} (R_{P,t} - \widehat{R}^{AR}_{P,t})^2}. \tag{8.8}$$

A positive R^2_{OOS} indicates a lower mean squared forecast error for the augmented model over the evaluated weeks.

Table 8.7: Out-of-Sample Forecasting Performance of Coverage

Measure	Diversified	Attacked + Security
Out-of-sample observations	172	172
RMSE, AR(6)	0.039699	0.043978
RMSE, AR(6) + Coverage	0.039229	0.043268
RMSE reduction (%)	1.19	1.61
Out-of-sample R^2 (%)	2.36	3.20
Clark–West statistic, HAC(6)	2.1149	2.4903
Clark–West p-value	**0.0172**	**0.0064**

Notes: One-week-ahead expanding-window forecasts begin after 234 observations. The benchmark includes six lags of weekly excess returns; the augmented model also includes six lags of $\Delta Coverage$. Clark–West p-values are one-sided and use HAC(6) standard errors. Bold denotes significance at the 5% level.

The augmented model achieves a 1.19% lower RMSE for Diversified and a 1.61% lower RMSE for Attacked + Security. The corresponding out-of-sample R^2 values are 2.36% and 3.20%. Clark–West p-values of 0.0172 and 0.0064 indicate a significant adjusted forecast-loss improvement under the

specified one-sided test. These p-values concern the Clark–West adjustment for nested models rather than a direct test of the raw RMSE difference.

The OOS RMSE exceeds the fitted residual RMSE reported in Table 8.6. The OOS figures measure errors on observations excluded from each estimation window and cover a different evaluation sample. Excluding 2020 reduces the OOS RMSE of the Coverage model to 0.02970 for Diversified and 0.03026 for Attacked + Security (119 forecast weeks remain). This calculation indicates that observations in 2020 contribute to the larger full-period OOS errors; the comparison also reflects the different estimation and evaluation samples.

Table 8.8: Out-of-Sample Forecasting Robustness Across Initial Estimation Windows

		Diversified		Attacked + Security	
Initial window	OOS forecasts	RMSE gain (%)	CW p-value	RMSE gain (%)	CW p-value
156	250	0.89	0.0244	1.22	0.0127
208	198	1.22	0.0148	1.64	0.0058
234	172	1.19	0.0172	1.61	0.0064
260	146	1.15	0.0216	1.62	0.0069

Notes: RMSE gain is the percentage reduction relative to the AR(6) benchmark. CW p-values are one-sided Clark–West tests using HAC(6) standard errors. Each initial window produces a different out-of-sample evaluation period. The four evaluation samples overlap substantially and are not independent replications.

The augmented model achieves lower RMSE in all four forecasting configurations. Clark–West p-values remain below 0.05 in both portfolios. Because the first forecast date changes with the initial estimation window, these comparisons measure sensitivity to the forecasting design across different evaluation periods. The overlapping forecast periods limit their interpretation as separate confirmations.

8.5 News-Based Indicators During Burst Episodes

The previous sections establish an association between lagged Coverage and subsequent weekly excess returns. This section examines weeks when cyber-risk reporting increases unusually sharply relative to its recent history.

The monthly Burst Detection procedure introduced in Section 6.4 is adapted to weekly observations. For each week, Coverage is compared with the mean and sample standard deviation of the preceding 52 weeks, excluding the current week. The standardized shock is

$$x_{N,t}^{(52)} = \frac{k_{N,t} - v_{N,t}^{(52)}}{w_{N,t}^{(52)}}, \tag{8.9}$$

where $k_{N,t}$ is normalized weekly Coverage and $v_{N,t}^{(52)}$ and $w_{N,t}^{(52)}$ are its trailing mean and sample standard deviation. The sample standard deviation is calculated with denominator $n - 1$, where n is the number of observations in the trailing window.

A Burst begins when the standardized shock exceeds three and current Coverage exceeds the 95th percentile of preceding observations. Once activated, the episode remains active while Coverage exceeds its rolling historical mean. Burst State identifies active weeks; Burst Intensity measures excess Coverage above that mean.

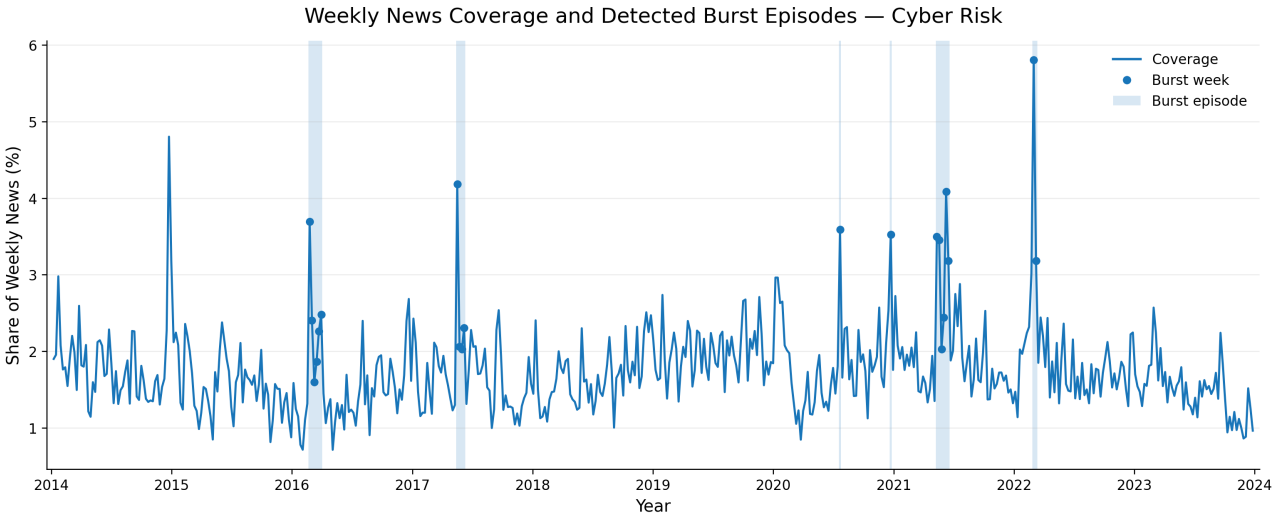


Figure 8.7: Weekly Cyber-Risk Coverage and detected Burst episodes, 2014–2023. The shaded intervals indicate weeks classified as Burst.

The first 52 weeks provide the historical reference window; elevated Coverage before that window is complete cannot trigger a Burst classification. The weekly procedure identifies six Burst episodes over 2014–2023. Their starting months are February 2016, May 2017, July 2020, December 2020, May 2021, and February 2022. After alignment with the common financial sample, 19 of the 366 regression observations are classified as Burst weeks.

8.5.1 Information characteristics during Burst episodes

Table 8.9: News-Based Indicators During Ordinary and Burst Weeks

Indicator	Ordinary weeks	Burst weeks	Difference
Coverage	0.01718	0.02835	+65.1%
Novelty	0.24316	0.23075	-5.1%
Semantic Drift	0.05953	0.08276	+39.0%
Observations	347	19	

Notes: The table reports indicator levels averaged across ordinary and Burst weeks in the common regression sample. Differences are expressed relative to the ordinary-week average. No formal test of differences is reported.

Coverage is higher during Burst weeks by construction, since exceptional Coverage is used to identify the episodes. Novelty averages 0.23075 during Burst weeks, compared with 0.24316 in ordinary weeks. Semantic Drift averages 0.08276 during Burst weeks, compared with 0.05953 in ordinary weeks. These differences describe the 19 Burst observations; their statistical significance is not assessed here.

8.5.2 Does financial relevance change during Burst periods?

A separate VAR(6) estimated only on Burst weeks would have too few observations to estimate its lag coefficients reliably. The 19 contemporaneous Burst observations also occur in separated episodes. The analysis therefore retains the common sample and uses a parsimonious interaction specification.

For indicator I, define the average of its preceding six weekly changes as

$$S_{I,t} = \frac{1}{6} \sum_{\ell=1}^{6} \Delta I_{t-\ell}. \tag{8.10}$$

The return equation is

$$\begin{aligned} R_{P,t} = \alpha_P + \sum_{\ell=1}^{6} \beta_{P,\ell} R_{P,t-\ell} + \gamma_P S_{I,t} \\ + \theta_P B_{t-1} + \delta_P (S_{I,t} B_{t-1}) + \varepsilon_{P,t}, \end{aligned} \tag{8.11}$$

where B_{t-1} indicates whether the preceding week was classified as Burst. The interaction coefficient δ_P measures the difference in the conditional association across the two states. Each indicator and portfolio is estimated separately, and the test concerns

$$H_0 : \delta_P = 0. \tag{8.12}$$

Table 8.10: Burst-Regime Interactions: HAC(6) Tests

Indicator	Diversified	Attacked + Security
Coverage	0.3132	0.2884
Novelty	0.3093	0.4294
Semantic Drift	0.8679	0.8231

Notes: Entries are HAC(6) p-values for the interaction between the mean of six lagged weekly indicator changes and the preceding week's Burst State. Each regression uses 366 observations. The common sample contains 19 contemporaneous Burst weeks and 20 observations with lagged Burst State equal to one. The lagged count differs because Burst State is shifted on the original weekly calendar before the 366 valid regression dates are selected.

For Coverage, the main-effect coefficient γ_P has HAC(6) p-values of 0.1081 for Diversified and 0.0895 for Attacked + Security. This coefficient describes the association when the preceding week is classified as ordinary. None of the interaction coefficients is significant at the 5% level. The small number of Burst observations limits the precision of these estimates, so the tests cannot establish that the associations are identical across states. Within the fitted interaction specifications, the data do not identify a statistically significant difference.

8.5.3 Coverage significance after excluding Burst weeks

The individual Coverage regression is also re-estimated after removing observations whose contemporaneous week is classified as Burst. The lagged variables retain their original weekly alignment, and the same HAC(6) joint test is applied.

Coverage remains significant in both portfolios after the 19 contemporaneous Burst observations are excluded. This comparison shows that the full-sample result persists among the retained financial observations. Coverage recorded during Burst episodes can still enter regressions for later non-Burst weeks through the six lagged regressors, so the restricted sample retains some information originating in Burst periods.

Table 8.11: Coverage Significance After Excluding Burst Weeks

Sample	Observations	Diversified	Attacked + Security
Full sample	366	**0.0076**	**0.0012**
Excluding Burst weeks	347	**0.0072**	**0.0005**

Notes: Entries are HAC(6) p-values for the joint significance of six lagged $\Delta Coverage$ coefficients in the $R_{P,t}$ equation. The restricted sample excludes 19 observations classified as Burst in the contemporaneous week. Regressors are constructed using the original chronology. Bold denotes significance at the 5% level.

8.6 To Sum Up

Coverage has the clearest statistical association with subsequent weekly portfolio excess returns among the three indicators examined. Its six lagged changes are jointly significant in both portfolios, after controlling for the other two indicators. Adding Coverage raises adjusted R^2 by 3.64 percentage points for Diversified and 4.43 percentage points for Attacked + Security.

The expanding-window exercise produces lower one-week-ahead forecast RMSE with Coverage in both portfolios. The reductions are 1.19% and 1.61%, with one-sided Clark–West HAC(6) p-values of 0.0172 and 0.0064. The alternative initial estimation windows also produce positive RMSE gains, although their evaluation periods overlap. The Burst interaction tests find no significant change in the estimated association across regimes, while the Coverage joint tests remain significant after contemporaneous Burst weeks are excluded.

These findings document a measurable predictive association in the specified weekly samples. The forecasting evidence concerns the portfolios and evaluation dates examined here.